

SLIDE 1

DESIGN AND IMPLEMENTATION OF A 32×32 INT8 SYSTOLIC ARRAY ACCELERATOR FOR TRANSFORMER GEMM ON KRIA KV260

FPGT Research Team

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Visuals

- Kria KV260 board
 - AI chip background
 - Team information
-

Presenter Script

Good morning respected lecturers and members of the evaluation committee.

Today, our team would like to present our research proposal entitled:

"Design and Implementation of a 32×32 INT8 Systolic Array Accelerator for Transformer GEMM on Kria KV260."

SLIDE 2

AI IS MOVING FROM CLOUD TO EDGE

The Rise of Large Language Models

- ChatGPT, Gemini, Llama and Qwen are transforming AI applications.
- AI is increasingly deployed in:
 - Autonomous vehicles
 - Smart cameras
 - Industrial IoT
 - Medical devices

Emerging Trend

Deploy AI directly on edge devices instead of relying entirely on cloud infrastructure.

Key Message

Edge AI is becoming a major direction for future AI deployment.

Visuals

4 large icons:

- ChatGPT
 - Self-driving car
 - Smart camera
 - Smart factory
-

Presenter Script

Large Language Models have rapidly become one of the most important technologies in modern AI.

However, instead of executing AI workloads entirely in cloud data centers, there is an increasing trend toward deploying AI directly on edge devices.

This trend is driven by the need for lower latency, better privacy, and reduced dependence on internet connectivity.

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WHAT DOES CURRENT RESEARCH SAY?

Recent Research Trends

Chitty-Venkata et al. (2023)

Transformer inference on edge devices is expected to become a key research direction in the near future.

Kang et al. (2024)

FPGAs provide:

- High parallelism
- Design flexibility
- Low power consumption

making them suitable for edge AI acceleration.

Li et al. (2023)

INT8 quantization significantly reduces computation and memory requirements.

Huang et al. (2023)

Systolic Array architectures demonstrate high efficiency for Transformer acceleration.

Conclusion

Recent studies indicate that FPGA-based Transformer acceleration is becoming an active research area.

Visuals

Timeline:

2023 → 2024

Chitty-Venkata

↓

Li

↓

Huang

↓

Kang

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CHALLENGES IN TRANSFORMER INFERENCE

Challenge 1

Massive Computation

Transformer models require enormous matrix operations.

Challenge 2

Memory Wall

Frequent transfers of large matrices between external memory and compute units create significant bandwidth bottlenecks.

Challenge 3

Complex Non-linear Functions

Softmax and GeLU require expensive computations and consume substantial hardware resources.

Visuals

Three large blocks:

 Computation

 Memory

 Non-linear Operations

Presenter Script

Although significant progress has been achieved, Transformer acceleration remains challenging.

Current systems still face three major issues:

high computational complexity,

memory bottlenecks,

and expensive non-linear operations.

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RESEARCH GAP

Existing Solutions

Most previous studies have focused on:

- High-performance FPGA platforms
 - Maximizing computational throughput
-

Remaining Problems

Gap 1

Communication bottlenecks between Processing System and Programmable Logic remain insufficiently addressed.

Gap 2

External memory bandwidth remains a major limitation.

Gap 3

Most studies target high-end FPGA boards such as:

- Xilinx Alveo U200
- Xilinx ZCU102

while limited work has focused on mid-range platforms such as Kria KV260.

Research Gap Statement

An efficient Transformer accelerator for resource-constrained SoC-FPGA platforms remains an open challenge.

Visuals

Table:

Existing Works	Limitation
High Throughput	High-end FPGA
Efficient Compute	Communication Bottleneck
Optimized Dataflow	Memory Bottleneck

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RESEARCH QUESTION

Main Research Question

How can a 32×32 Systolic Array architecture be designed on the Kria KV260 SoC platform to achieve an effective balance among:

- Throughput
- Latency
- Hardware Resource Utilization

for Transformer GEMM acceleration?

Visuals

Large question mark in center.

Three surrounding circles:

Throughput

Latency

Resources

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PROPOSED SOLUTION

Hardware / Software Co-design

Instead of implementing the entire Transformer on FPGA:

CPU (PS)

Handles:

- Softmax
 - GeLU
 - Control Logic
-

FPGA (PL)

Accelerates:

- General Matrix Multiplication (GEMM)
-

Key Techniques

✓ 32×32 Systolic Array

✓ INT8 Quantization

✓ Matrix Tiling

Visuals

CPU → Softmax / GeLU

↓

FPGA → GEMM Accelerator

Key Message

Focus FPGA resources exclusively on the main computational bottleneck.

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SYSTEM ARCHITECTURE & EVALUATION

Hardware Platform

Xilinx Kria KV260 SoC

Benchmark Models

- Qwen
 - Llama
 - Gemma
-

Evaluation Metrics

- Latency
 - Throughput (GOPS)
 - DSP Utilization
 - LUT Utilization
 - BRAM Utilization
-

Visuals

Use simplified Figure 2 from proposal.

Avoid showing AXI signal names.

Only show:

CPU ↔ DMA ↔ Accelerator ↔ Memory

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EXPECTED CONTRIBUTIONS

Academic Contribution

- Demonstrate an efficient Transformer acceleration methodology for Edge AI systems.
-

Engineering Contribution

- Develop a complete GEMM accelerator IP on Kria KV260.
-

Practical Applications

- Industrial IoT
 - Autonomous Vehicles
 - Medical Devices
 - On-device AI Assistants
-

Visuals

4 application icons.

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CONCLUSION

Key Takeaways

- ✓ Edge AI is becoming increasingly important.
 - ✓ Transformer inference remains computationally expensive.
 - ✓ Existing solutions still face communication and memory bottlenecks.
 - ✓ Our research proposes a Hardware/Software Co-design architecture based on a 32×32 INT8 Systolic Array on Kria KV260.
 - ✓ The proposed solution aims to improve performance while maintaining resource efficiency on embedded FPGA platforms.
-

Final Message

"Efficient Transformer acceleration on resource-constrained Edge AI platforms."

Script

SLIDE 1

Hello teacher

Thank you for your time.

We are the FPGT research team.

Today, we would like to present our research

Our work focuses on accelerating Transformer inference on an FPGA-based embedded platform.

SLIDE 2

In recent years, Large Language Models such as ChatGPT, Gemini, Llama have become very popular.

But Most of these models run on cloud servers.

They need send data to the cloud and may have privacy concerns.

Because of that, many companies are trying to run AI models directly on edge devices.

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To understand this area, we reviewed several recent studies.

First, Chitty-Venkata and his team showed that running Transformer models on edge devices is becoming an important research direction.

Next, Kang and his team found that FPGA platforms are suitable for AI acceleration because they provide high parallelism and low power consumption.

Other researchers focused on optimization techniques.

For example, Li and his team used INT8 quantization to reduce computation and memory requirements.

Huang and his team used Systolic Arrays to improve Transformer performance.

From these studies, we can see that FPGA-based Transformer acceleration is receiving a lot of attention.

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Although many researchers have worked on this topic, several challenges still remain.

The first challenge is computation.

Transformer models perform a large number of matrix multiplication operations.

The second challenge is memory access.

The system needs to move a large amount of data between memory and processing units.

This often becomes a bottleneck.

The third challenge comes from functions such as Softmax and GeLU.

These functions are difficult to implement efficiently on FPGA hardware.

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After reviewing the literature, we found several research gaps.

First, many studies focus on improving computational performance.

However, communication between the CPU and FPGA still creates overhead.

Second, memory bandwidth remains a limitation.

Third, many previous studies used powerful FPGA boards such as Alveo U200 and ZCU102.

However, fewer studies have focused on mid-range platforms such as the Kria KV260.

Because of these limitations, efficient Transformer acceleration on embedded FPGA systems is still an open problem.

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So, based on everything we just saw, we asked ourselves one main question.

How can we actually build a 32-by-32 Systolic Array on the Kria KV260 board and still get a good balance between speed, delay, and hardware space?

Basically, this single question drives our entire design.

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To answer that, we went with a Hardware/Software Co-design approach.

Simply put, we split the work between the CPU and the FPGA.

The CPU takes care of the tricky non-linear stuff, like Softmax and GeLU.

Meanwhile, the FPGA is fully dedicated to GEMM operations.

Why? Because GEMM is the biggest bottleneck when running Transformers.

To make this work, we combined three things.

We used a 32-by-32 Systolic Array, INT8 Quantization, and Matrix Tiling.

Together, they really boost performance and cut down on memory traffic.

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Now, for the actual setup, we are putting this design on the Kria KV260 platform.

To see how well it works, we will test it with some popular compact language models, like Qwen, Llama, and Gemma.

We want to measure a few key things like Latency, Throughput, and how much DSP, LUT, and BRAM we are using.

Also, to make sure it is doing the math right, we will double-check our hardware outputs against PyTorch.

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Looking ahead, we hope this project pays off in three ways.

First off, we want to build a working GEMM accelerator and prove it works on real hardware.

Second, we are planning to share what we learned.

We are aiming to submit a paper to FISAT, so that other people can see our methods and results.

And lastly, in the real world, this kind of design could be super useful for things like IoT, self-driving cars, or medical devices.

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So, just to wrap things up.

It is pretty clear that there is a huge demand for running Large Language Models on edge devices right now.

But, it is still really hard to do because of the heavy math and memory limits.

That is exactly what our research is trying to fix.

By using a Co-design approach and a 32-by-32 INT8 array on the Kria board, we hope to offer a really practical way to run Transformers on smaller FPGA systems.

And that brings us to the end of our talk.

Thank you so much for listening.

We would love to hear any questions or thoughts you might have.